

Binary Search: Finding a Book in an Alphabetized Bookshelf

Let's say you have a bookshelf of 100 books. You can't remember which books you have or their exact locations on the shelf, but you do know that they are sorted alphabetically by title. To find your book *Zebras: The Complete Guide*, you wouldn't start at the beginning of the bookshelf, where *Aaron Burr Biography* is, but rather toward the end of the bookshelf. Your zebra book wouldn't be the very last book on the shelf if you also had books on zephyrs, zoos, and zygotes, but it would be close. Thus, you can use the facts that the books are in alphabetical order and that Z is the last letter of the alphabet as *heuristics*, or approximate clues, to look toward the end of the shelf rather than the beginning.

Binary search is a technique for locating a target item in a sorted list by repeatedly determining which half of the list the item is in. The most impartial way to search the bookshelf is to start with a book in the middle, and then ascertain if the target book you're looking for is in the left half or the right half.

You can then repeat this process, as shown in [Figure 5-1](#): look at the book in the middle of your chosen half and then determine whether your target book is in the left-side quarter or the right-side quarter. You can do this until you either find the book, or find the place where the book should be but isn't and declare that the book doesn't exist on the shelf.